# EXHAUSTIVE SEARCHES FOR WILLIAMS NUMBERS: THE FOUR-PRIME CASE AND A GENERAL $10^{24}$ BOUND

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Abstract. An odd squarefree composite integer is a Williams number if it is both Carmichael and Lucas–Carmichael; none is known. For a four-prime Williams number $n = p_1p_2p_3p_4$ we give a self-contained restatement, with a slight sharpening, of the bounds due to McIntosh and Dipra, and derive from them an enumeration that fixes $(p_1, p_2)$, places $p_3p_4$ in a single residue class modulo $\operatorname{lcm}\big((p_1^2-1)/2, (p_2^2-1)/2\big)$, and rejects almost every candidate with one multiplication. Certified computations prove that no such number satisfies $n \le 10^{30}$ or $p_1 \le 500{,}000$, extending McIntosh and Dipra's $n > 10^{24}$, $p_1 > 2\times10^4$. Separately, a factor-and-test scan of all 308,279,939 Carmichael numbers below $10^{24}$ finds no Williams number. Code, logs and certification records are archived.

## 1. Introduction

Throughout, $n$ denotes an odd squarefree composite integer with prime factorisation $n = p_1p_2\cdots p_k$, $p_1 < \cdots < p_k$, $k = \omega(n) \ge 2$.

**Definition 1.1.** $n$ is a *Carmichael number* if $(p-1) \mid (n-1)$ for every $p \mid n$ (Korselt's criterion), and a *Lucas–Carmichael number* if $(p+1) \mid (n+1)$ for every $p \mid n$. A number that is both is a *Williams number* (a 1-Williams number in Echi's terminology [9]).

Carmichael numbers are the absolute Fermat pseudoprimes; Lucas–Carmichael numbers play the same role for Lucas sequences with discriminant a quadratic non-residue of every $p \mid n$. H. C. Williams [27] asked in 1977 whether a number can be both, proved that it cannot have exactly three prime factors and cannot be divisible by 3, and showed that under his Lucas-sequence formulation the number of prime factors must be odd (in the divisibility formulation above, all $p_i$ lie in one class modulo 4, and $k$ is odd when that class is 3; Lemma 2.2). Echi [9] named such numbers Williams numbers, within a family of $a$-Williams numbers, and dropped the parity restriction; Bouallègue, Echi and Pinch [4] and Al-Rasasi and Ghanmi [1] studied general $a$; Pinch [20] reported that there is no Williams number up to $10^{16}$ with an odd number of prime factors. McIntosh [17] treated the parallel problem $(p \pm 1) \mid (n-1)$, and McIntosh and Dipra [18] the Williams numbers themselves. Zhang [28] developed counting heuristics for Williams numbers built from a fixed set of primes and offered a prize for the first example. No Williams number is known.

The question is tied to primality testing. Pomerance's 1984 heuristic argument that counterexamples to the Baillie–PSW primality test exist [22], restated in [5], builds $n = \prod p$ from primes with $p \equiv 3 \pmod 8$, $(5/p) = -1$ and smooth squarefree $(p-1)/2$, $(p+1)/4$, chosen so that $(p-1) \mid (n-1)$ and $(p+1) \mid (n+1)$ for all $p \mid n$; such an $n$ is in particular a Williams number [17, 18, 28]. The construction attempted by Chen and Greene [6] for Grantham's \$620 problem [13] uses weaker order conditions; the related set of 2,030 primes does satisfy $(p-1) \mid M$,

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$(p+1) \mid N$, and an unrefereed report by Phatak [19] states that no subset of that set is a Williams number. The Lenstra–Pomerance analysis of Agrawal's conjecture from the AKS primality test [2, 16] shows that the natural counterexamples are Williams numbers whose primes are all $\equiv 3 \pmod{80}$. Howe's higher-order Carmichael numbers [14] are a further neighbour: by Howe's criterion a squarefree $n$ is Carmichael of order 2 iff for every $p \mid n$ one has $n \equiv 1$ or $n \equiv p \pmod{p^2-1}$, while $n$ is a Williams number iff $n \equiv p \pmod{(p^2-1)/2}$ for every $p \mid n$ (Lemma 2.1).

Prior results for four prime factors, and what is new here. McIntosh and Dipra [18] proved that for a fixed set of $d-3$ prime factors only finitely many Williams numbers with $d$ prime factors exist, with explicit bounds on the remaining three primes, observed that all prime factors are congruent modulo 12 (and $\equiv 1$ when their number is even), organised two searches, and showed that a Williams number with exactly four prime factors has $p_1 > 2 \times 10^4$ and $n > 10^{24}$. Table 1 sets their results for $d = 4$ beside ours. Theorem 3.1 is their theorem restated with a self-contained proof; the new contributions are the enumeration algorithm of §4, its certified completeness (§5), the resulting exclusions (Theorem 6.1), and the general $10^{24}$ census (Theorem 7.1).

| | McIntosh–Dipra [18] | this paper |
|---|---|---|
| all $p_i \equiv 1$ (12) | §2 | Lemma 2.2(c) |
| $p_2$ | $p_2 < 5p_1^{3/2}$ (Thm. 3) | $p_2^2 \le 24p_1^3 + 1$ (Thm. 3.1) |
| $p_3$ | $p_3 < p_1p_2$ (Thm. 3) | $p_3 < 2p_1p_2$ (weaker; sufficient) |
| $p_4$ | $p_4 < \sqrt{2p_1p_2p_3}$ (Thm. 3) | same |
| search method | loops over the three quotients; Pinch-style loops over compatible primes | CRT class for $p_3p_4$ and window filter (§4); certified completeness |
| four primes excluded | $p_1 \le 2 \times 10^4$; $n \le 10^{24}$ | $p_1 \le 500{,}000$; $n \le 10^{30}$ (Thm. 6.1) |
| all $d$ | searches for $4 \le d \le 13$ at unstated bounds in $[10^{20}, 10^{35}]$ | none below $10^{24}$, any $d$ (Thm. 7.1) |

TABLE 1. Prior and present results: four-prime Williams numbers (rows 1–6) and all Williams numbers (last row).

All code, logs and certification records are available (§9). Appendix A proves that Chernick-type polynomial families whose divisibilities hold identically in the parameter cannot produce Williams numbers with three factors, with uniform offsets $\pm 1$, or in generic position; an unpublished exploratory note in the repository [26] records a measurement of the two-sided subset-product method; none of its conclusions is used here.

## 2. PRELIMINARIES

For an odd prime $p$ write $L_p = (p^2-1)/2 = \operatorname{lcm}(p-1, p+1)$.

**Lemma 2.1.** *Let $p \mid n$. Then $(p-1) \mid (n-1)$ and $(p+1) \mid (n+1)$ hold together if and only if $n/p \equiv 1 \pmod{L_p}$, equivalently $n \equiv p \pmod{L_p}$.*

*Proof.* The two divisibilities say $n \equiv 1 \pmod{p-1}$ and $n \equiv -1 \pmod{p+1}$. Since $p \equiv 1 \pmod{p-1}$ and $p \equiv -1 \pmod{p+1}$, this is $n \equiv p$ modulo $\operatorname{lcm}(p-1, p+1) = L_p$ (the gcd of $p \mp 1$ is 2). Finally $L_p \mid n - p = p\,(n/p - 1)$ with $\gcd(p, L_p) = 1$ gives $L_p \mid n/p - 1$, and conversely. □

**Lemma 2.2** (congruence classes)**.** *Let $n$ be a Williams number with $k$ prime factors.*

(a) *All $p \mid n$ lie in one class $c$ modulo 4, and $n \equiv c \pmod 4$; hence $k$ is odd if $c = 3$.*
(b) *$3 \nmid n$; all $p \mid n$ lie in one class $c'$ modulo 3, and $n \equiv c' \pmod 3$; hence $k$ is odd if $c' = 2$.*
(c) *If $k$ is even, every $p \mid n$ is $\equiv 1 \pmod{12}$.*

*Proof.* (a) $8 \mid p^2 - 1$ for odd $p$, so $4 \mid L_p$ and $n \equiv p \pmod 4$ for every $p \mid n$ by Lemma 2.1. Thus all $p \equiv c$, and $n \equiv c^k \pmod 4$; if $c = 3$ this forces $k$ odd. (b) For $p \neq 3$, $3 \mid p^2 - 1$, so $3 \mid L_p$ and $n \equiv p \pmod 3$. If $3 \mid n$ then every other prime factor would be $\equiv n \equiv 0 \pmod 3$, impossible for $k \geq 2$; so $3 \nmid n$, all $p \equiv c' \pmod 3$ and $n \equiv c'^k \equiv c'$, which for $c' = 2$ forces $k$ odd. (c) is (a) and (b) with $k$ even. □

Part (c) is observed in [18, §2]. Williams proved that $k = 3$ is impossible; the following proof is essentially that of [18, Thm. 1].

**Proposition 2.3** (Williams)**.** *No Williams number has exactly three prime factors.*

*Proof.* Let $n = p_1p_2p_3$. By Lemma 2.1 at $p_3$, $p_1p_2 - 1 = tL_3$ with $t \geq 1$ (as $p_1p_2 > 1$); since $p_1p_2 < p_3^2$ we have $t < 2$, so $2p_1p_2 = p_3^2 + 1$. At $p_2$, $p_1p_3 - 1 = sL_2$ with $s \geq 1$; now $p_3 < \sqrt{2p_1p_2} < \sqrt{2}\,p_2$, so $p_1p_3 < \sqrt{2}\,p_2^2$ and $s < 2\sqrt{2}\,p_2^2/(p_2^2 - 1) < 3$ (as $p_2 \geq 5$). If $s = 2$ then $p_1p_3 = p_2^2$, impossible; so $2p_1p_3 = p_2^2 + 1$. Subtracting, $2p_1(p_2 - p_3) = p_3^2 - p_2^2$, i.e. $2p_1 = -(p_2 + p_3) < 0$, a contradiction. □

## 3. Four prime factors: structural bounds

**Theorem 3.1** (McIntosh–Dipra [18], $d = 4$; restated)**.** *Let $n = p_1p_2p_3p_4$, $p_1 < p_2 < p_3 < p_4$, be a Williams number, and write $P = p_1p_2$, $q = p_3$, $r = p_4$.*

(a) *Every $p_i \equiv 1 \pmod{12}$; in particular $p_1 \geq 13$ and $p_2 \geq 37$.*
(b) *There are integers $d, e, s \geq 1$ with*

$$2Pq = d(r^2 - 1) + 2, \qquad 2Pr = e(q^2 - 1) + 2, \qquad 2p_1qr = s(p_2^2 - 1) + 2, \tag{1}$$

*and consequently $q < 2P$, $r < \sqrt{2Pq} < 2P$, $m := qr < 4P^2$, $n < 4P^3$ and $d < 2p_1$.*
(c) *$p_2^2 \leq 24p_1^3 + 1$.*

*Consequently $p_2 < 4.9\,p_1^{3/2}$, $p_3, p_4 < 2P < 9.8\,p_1^{5/2}$ and $n < 477\,p_1^{15/2}$.*

*Proof.* (a) is Lemma 2.2(c).

(b) Lemma 2.1 at $r$ gives $n/r - 1 = Pq - 1 = dL_r$ with an integer $d \geq 1$, which is the first equation; the second and third are Lemma 2.1 at $q$ and at $p_2$. From $2Pq = d(r^2-1)+2 \geq r^2+1 > q^2$ we get $q < 2P$ and $r^2 < 2Pq$, hence $r < \sqrt{2Pq} < 2P$, $m = qr < 4P^2$ and $n = Pm < 4P^3$. Also $d = 2(Pq - 1)/(r^2 - 1) < 2Pq/(r^2 - 1) < 2P/q < 2p_1$ because $r \geq q + 1$ and $q > p_2$.

(c) Multiply the three equations in (1). The left side is $8P^2p_1\,q^2r^2 = 8p_1^3p_2^2q^2r^2$. Writing $d(r^2-1)+2 = dr^2\big(1-\frac{d-2}{dr^2}\big)$ and similarly for the other two factors, the right side is $d\,e\,s\;r^2q^2p_2^2\,\Pi$ with

$$\Pi = \Big(1 - \frac{d-2}{dr^2}\Big)\Big(1 - \frac{e-2}{eq^2}\Big)\Big(1 - \frac{s-2}{sp_2^2}\Big).$$

Hence

$$d\,e\,s \cdot \Pi = 8p_1^3. \tag{2}$$

For an integer $t \geq 1$ and real $x > 0$ one has $-1/x^2 \leq (t-2)/(tx^2) < 1/x^2$, so each factor of $\Pi$ lies in $[1 - 1/x^2,\ 1 + 1/x^2]$ with $x \in \{r, q, p_2\}$, $x \geq p_2$; therefore

$$(1 - p_2^{-2})^3 \leq \Pi \leq (1 + p_2^{-2})^3. \tag{3}$$

Now reduce the three equations (1) modulo $p_1$. Each left side is divisible by $p_1$, so $d(r^2 - 1) \equiv e(q^2 - 1) \equiv s(p_2^2 - 1) \equiv -2 \pmod{p_1}$. As $p_1$ is odd, $p_1 \nmid d$, $p_1 \nmid e$ and $p_1 \nmid s$. Hence $des \neq 8p_1^3$, i.e. $\Pi \neq 1$.

*Case* $\Pi < 1$. Then $des \geq 8p_1^3+1$ and $\Pi = 8p_1^3/des \leq 1-\frac{1}{8p_1^3+1}$; with $\Pi \geq (1-p_2^{-2})^3 \geq 1-3p_2^{-2}$ this gives $p_2^2 \leq 3(8p_1^3+1) = 24p_1^3+3$. Since $p_2 \equiv 1 \pmod{12}$ implies $p_2^2 \equiv 1 \pmod{24}$, while $24p_1^3+2$ and $24p_1^3+3$ are $\equiv 2, 3 \pmod{24}$, in fact $p_2^2 \leq 24p_1^3+1$.

*Case* $\Pi > 1$. A factor $1-\frac{t-2}{tx^2}$ of $\Pi$ exceeds 1 only when $t=1$. Now $e > d \geq 1$, because $e/d = \frac{(Pr-1)(r^2-1)}{(Pq-1)(q^2-1)} > 1$ for $r > q$; and $s > 2p_1$, because $qr \geq p_2^2+1$ gives $s = \frac{2(p_1qr-1)}{p_2^2-1} \geq \frac{2(p_1p_2^2+p_1-1)}{p_2^2-1} > 2p_1$. So only the first factor can exceed 1, and $\Pi \leq 1+r^{-2}$. On the other hand $des \leq 8p_1^3-1$ gives $\Pi = 8p_1^3/des \geq 1+\frac{1}{8p_1^3-1}$. Hence $r^2 \leq 8p_1^3-1$, and $p_2^2 < r^2 < 8p_1^3 \leq 24p_1^3+1$.

In both cases $p_2^2 \leq 24p_1^3+1$. The consequences follow from (b) and (c): $p_2^2 \leq 24p_1^3+1 < 24.01\,p_1^3 = (4.9\,p_1^{3/2})^2$ for $p_1 \geq 5$, so $p_2 < 4.9\,p_1^{3/2}$; then $q, r < 2P < 9.8\,p_1^{5/2}$ and $n < 4P^3 < 4 \cdot 4.9^3\,p_1^{15/2} < 477\,p_1^{15/2}$. □

*Remark* 3.2. In the notation of [18] ($P, q, r, s$ and $f_s, f_r, f_q$ for our $p_1, p_2, p_3, p_4$ and $d, e, s$), Theorems 2 and 3 there give $q < 5P^{3/2}$, $r < Pq$ and $s < \sqrt{2Pqr}$, and show by Newton's inequality that the quotient product always exceeds $8P^3$, so that the case $\Pi > 1$ above never occurs. The statement above differs only in the slightly sharper bound on $p_2$ (from the congruence modulo 24) and the weaker bound on $p_3$, which is all the enumeration needs.

**Corollary 3.3.** *For each prime $p_1$ there are only finitely many four-prime Williams numbers with smallest prime $p_1$, all below $477\,p_1^{15/2}$. Conversely a four-prime Williams number $n$ has $p_1 > (n/477)^{2/15}$.*

## 4. The enumeration algorithm

Theorem 3.1 makes two kinds of complete enumeration possible: all $n \leq B$ (where $n > p_1p_2^3$ forces $p_1 < B^{1/4}$ and $p_2 < (B/p_1)^{1/3}$), and all $n$ whatsoever with $p_1 \leq X$ (where $p_2 \leq \sqrt{24p_1^3+1}$). Both reduce to a search over pairs $(p_1, p_2)$ of primes $\equiv 1 \pmod{12}$.

**Lemma 4.1** (one class for the cofactor)**.** *With notation as in Theorem 3.1, $m = qr$ satisfies $p_2m \equiv 1 \pmod{L_{p_1}}$ and $p_1m \equiv 1 \pmod{L_{p_2}}$. These are solvable iff $p_1 \nmid p_2^2-1$ and $p_1 \equiv p_2 \pmod{\gcd(L_{p_1}, L_{p_2})}$, in which case $m$ lies in a single residue class modulo $\Lambda = \operatorname{lcm}(L_{p_1}, L_{p_2})$.*

*Proof.* The congruences are Lemma 2.1 at $p_1$ and $p_2$ ($n/p_1 = p_2m$, $n/p_2 = p_1m$). The first is always solvable: $p_2 \mid L_{p_1}$ would need $p_2 \mid p_1 \pm 1$ with $p_2 > p_1$, impossible for odd primes. The second is solvable iff $\gcd(p_1, L_{p_2}) = 1$, i.e. $p_1 \nmid p_2^2-1$. Given both, let $g = \gcd(L_{p_1}, L_{p_2})$; the pair is compatible iff $p_2^{-1} \equiv p_1^{-1} \pmod{g}$, i.e. $p_1 \equiv p_2 \pmod{g}$ (note $\gcd(p_1p_2, g) = 1$). The rest is the Chinese remainder theorem. □

Since $\Lambda \approx P^2/(4g)$ while $m < 4P^2$, Lemma 4.1 leaves $O(g)$ candidates $m$ per pair. The next lemma rejects almost all of them with one multiplication.

**Lemma 4.2** (window)**.** *With $d, e$ as in (1) and $j := de$, $T := 4P^2$,*

$$|jm - T| < \max\bigl(2j\sqrt{2p_1},\ 8p_1\bigr). \tag{4}$$

*Moreover, for fixed $m > 2\lceil\sqrt{2p_1}\rceil$ (which holds for $m > p_2^2$) the set of real $j > 0$ satisfying $|jm-T| \leq \max(2j\lceil\sqrt{2p_1}\rceil, 8p_1)$ is an interval containing $T/m$; hence it contains an integer iff it contains $\lfloor T/m \rfloor$ or $\lceil T/m \rceil$.*

*Proof.* From (1), $d = 2(Pq-1)/(r^2-1)$ and $e = 2(Pr-1)/(q^2-1)$, so $j\,(q^2-1)(r^2-1) = 4(Pq-1)(Pr-1)$. Expanding with $m = qr$,

$$jm^2 - Tm = j(q^2+r^2-1) - 4P(q+r) + 4.$$

The right side is a difference of two positive numbers, so its absolute value is at most $\max\big(j(q^2+r^2-1),\ 4P(q+r)-4\big) \le \max(2jr^2,\ 8Pr)$. Dividing by $m = qr$ gives $|jm-T| \le \max(2jr/q,\ 8P/q)$. By Theorem 3.1(b), $r^2 < 2Pq$, so $r/q < \sqrt{2P/q} < \sqrt{2p_1}$ (as $q > p_2$), and $P/q < p_1$; this is (4). For the interval property, put $S = \lceil\sqrt{2p_1}\rceil$ and $\phi(j) = |jm - T| - \max(2jS, 8p_1)$. On $j \ge T/m$ the first term increases with slope $m$ and the second with slope at most $2S < m$, so $\phi$ is strictly increasing; on $j \le T/m$ the first term decreases and the second is non-decreasing, so $\phi$ is strictly decreasing. Thus $\{\phi \le 0\}$ is an interval, and it contains $T/m$ since $\phi(T/m) < 0$. □

**Lemma 4.3** (reconstruction)**.** *Let $n = p_1p_2qr$ be a four-prime Williams number with $d$ as in (1). Then $r$ is a root of $dx^3 - (d-2)x - 2n$, and $\lfloor(2n/d)^{1/3}\rfloor \in \{r-1, r\}$.*

*Proof.* Multiplying the first equation of (1) by $r$ gives $2n = 2Pqr = dr^3-(d-2)r$. Put $x = 2n/d$; then $x - r^3 = -(d-2)r/d \in (-r, r]$. If $x \ge r^3$ then $r \le x^{1/3} < (r^3+r)^{1/3} < r+1$; if $x < r^3$ then $x > r^3 - r > (r-1)^3$ (as $3r^2 - 3r + 1 > r$ for $r \ge 1$), so $r - 1 < x^{1/3} < r$. Thus $\lfloor x^{1/3}\rfloor$ is $r$ or $r-1$, and the same holds for $\lfloor\lfloor x\rfloor^{1/3}\rfloor$ since $\lfloor x\rfloor \le x < \lfloor x\rfloor + 1$ changes the cube root by less than 1 in the same direction. □

**Algorithm 1** Enumeration of four-prime Williams numbers (campaign 1: all $n \le B$; campaign 2: all $n$ with $p_1 \le X$)

**Input:** $B$ (campaign 1) or $X$ (campaign 2). Put $S = \lceil\sqrt{2p_1}\rceil$, $R_j = \max(2jS, 8p_1)$; call $(m, j)$ *admissible* if $|jm - T| \le R_j$ where $T = 4P^2$.
**for all** primes $p_1 \equiv 1\ (12)$ with $p_1 < B^{1/4}$ (campaign 1) or $p_1 \le X$ (campaign 2) **do**
  **for all** primes $p_2 \equiv 1\ (12)$, $p_1 < p_2 \le \sqrt{24p_1^3+1}$, and $p_2 < (B/p_1)^{1/3}$ in campaign 1 **do**
  ▷ the implementation uses $\sqrt{24p_1^3+3}$, equivalent by the congruence modulo 24
    $P \leftarrow p_1p_2$; **if** $p_1 \mid p_2^2 - 1$ or $p_1 \not\equiv p_2 \pmod{\gcd(L_{p_1}, L_{p_2})}$ **then skip** (Lemma 4.1)
    compute $m_0, \Lambda$ by CRT; $m_{\max} \leftarrow \min(4P^2 - 1, \lfloor B/P\rfloor)$ (no second term in campaign 2)
    $M^* \leftarrow \min\big(m_{\max}, \lfloor\sqrt{T}\sqrt{\Lambda}\rfloor\big)$ ▷ any $M^*$ with $2S < M^* \le m_{\max}$ is valid; this choice balances the two walks; if $M^* = m_{\max}$ walk B is empty
    **for all** $m \equiv m_0\ (\Lambda)$ with $p_2^2 < m \le \min(M^*, m_{\max})$ **do** ▷ walk A
      keep $m$ iff $(m, \lfloor T/m\rfloor)$ or $(m, \lceil T/m\rceil)$ is admissible
    **end for**
    **for all** integers $j$ from $\max\big(1, \lfloor(T-8p_1)/(m_{\max}+2S)\rfloor - 1\big)$ to $\lfloor(T+8p_1)/(M^*-2S)\rfloor + 1$ **do** ▷ walk B
      keep every $m \equiv m_0\ (\Lambda)$ with $\max(M^*+1, \lceil(T-R_j)/j\rceil) \le m \le \min(m_{\max}, \lfloor(T+R_j)/j\rfloor)$, unless $(m, j-1)$ is admissible (reported once, under the least admissible $j$)
    **end for**
    **for all** kept $m$ **do** ▷ survivors
      $n \leftarrow Pm$; **if** $2^{n-1} \not\equiv 1 \pmod n$ **then discard** (in C with 128-bit arithmetic when $P \le 2^{40}$; otherwise deferred to the Python verifier)
      **for all** $d = 1, \ldots, 2p_1$ **do**
        $x \leftarrow \lfloor\lfloor 2n/d\rfloor^{1/3}\rfloor$; **for** $r \in \{x-1, x, x+1\}$ with $dr^3 - (d-2)r = 2n$ and $r \mid m$: $q \leftarrow m/r$; report $(p_1, p_2, q, r)$ if $p_2 < q < r$, $q, r$ prime and all four conditions of Lemma 2.1 hold
      **end for**
    **end for**
  **end for**
**end for**

## 5. Completeness, validation and certification

**Proposition 5.1** (completeness)**.** *Let $n = p_1p_2qr$ be a four-prime Williams number with $n \le B$ (campaign 1) or $p_1 \le X$ (campaign 2). Then Algorithm 1 visits the pair $(p_1, p_2)$, keeps $m = qr$, and reports $n$.*

*Proof.* By Theorem 3.1(a) both $p_1, p_2$ are primes $\equiv 1 \pmod{12}$. If $n \le B$ then $p_1^4 < n \le B$ and $p_1p_2^3 < n \le B$, so $p_1 < B^{1/4}$ and $p_2 < (B/p_1)^{1/3}$; in either campaign $p_2 \le \sqrt{24p_1^3+1}$ by Theorem 3.1(c). Hence the pair is in the range scanned. The congruences of Lemma 4.1 hold for $m$, so the pair is not skipped and $m \equiv m_0 \pmod{\Lambda}$; by Theorem 3.1(b), $p_2^2 < m < 4P^2$, and $m \le B/P$ when $n \le B$, so $m$ lies in the walked range. By Lemma 4.2 the pair $(m, de)$ is admissible. If $m \le M^*$ then $\lfloor T/m \rfloor$ or $\lceil T/m \rceil$ is admissible and walk A keeps $m$. If $m > M^*$ then $j = de$ lies in the scanned range, because admissibility gives $jm \le T + 2jS + 8p_1$ and $jm \ge T - 2jS - 8p_1$, hence $j(m-2S) \le T + 8p_1$ with $m - 2S > M^* - 2S$ and $j(m+2S) \ge T - 8p_1$ with $m + 2S \le m_{\max} + 2S$; and $m$ lies in the exact window for the least admissible $j$, so walk B keeps it. Since $n$ is a Carmichael number, $2^{n-1} \equiv 1 \pmod{n}$. Finally $1 \le d < 2p_1$ by Theorem 3.1(b), and by Lemma 4.3 the integer cube root lands within one of $r$, so the loop over $d$ recovers $r$, $q = m/r$, and all checks pass. □

Validation. The implementation was tested against brute force on a *relaxed* problem that has solutions: replace $L_p$ by $(p^2-1)/\kappa$ with $\kappa \in \{24, 120\}$ (Theorem 3.1(b) and Lemmas 4.1–4.3, which are all the algorithm uses, go through with 2 replaced by $\kappa$; Theorem 3.1(c) and the Fermat filter are specific to $\kappa = 2$ and are switched off in these runs). On the relaxed problem the enumerator finds exactly the solution set that a direct four-fold loop finds; for $\kappa = 24$ the unique solution below $10^{12}$ is $13685 = 5 \cdot 7 \cdot 17 \cdot 23$, and the outputs of both programs are archived. Separate self-tests brute-force the two-prime sub-problem $(q, r)$ for small $(p_1, p_2)$ in four relaxed variants and check every inequality of Lemmas 4.2–4.3 on the solutions found, brute-force the residue class of Lemma 4.1 for 13,037 pairs, and compare the 128-bit Fermat routine with Python on random moduli. The two walks were checked to produce identical survivor lists on six instances, two of them relaxed instances with 346,409 and 11,007,875 survivors. These are regression tests of the implementation on instances with known answers, not an independent validation of the production campaigns, which rests on Proposition 5.1 and the certification below.

Certification. Each campaign is processed in blocks of consecutive $p_1$ and logs one record per block (its $p_1$ range, number of pairs, survivors, Fermat survivors, solutions). Both campaigns were run twice, the second time with every window survivor written out, and a certification script asserts: every block identifier appears exactly once; block ranges are increasing, disjoint and contiguous over the admissible primes; per-block counts equal those of the first run; the distinct survivors, within each campaign, number exactly the logged total, block by block; every survivor re-tested in Python (base-2 Fermat test on $n = Pm$, then factorisation of $m$ and the four conditions of Lemma 2.1 on the original tuple) yields the logged number of Fermat survivors and no Williams number. The records include hashes of the binary, the source and the logs, the compiler, host and source revision. The certifications were repeated from a clean tagged checkout in an independent AI-based review process (see the declaration below), and the complete test suite was run from a clean archive of the tagged source on a freshly created cloud machine (transcript in the repository).

## 6. Computational results for four prime factors

**Theorem 6.1.** *There is no Williams number with exactly four prime factors and $n \le 10^{30}$, and there is none at all whose smallest prime factor is at most* 500,000*. Consequently a four-prime*

*Williams number, if one exists, has* $p_1 > 500{,}000$ *and* $n > 10^{30}$. *(McIntosh and Dipra* [18] *had* $p_1 > 2 \times 10^4$ *and* $n > 10^{24}$.*)*

*Computation.* Table 2 summarises the two campaigns (Apple M4 Pro, 14 threads). In each, the single survivor of the Fermat filter is the Carmichael number $26921089 = 13 \cdot 37 \cdot 97 \cdot 577$, for which $38 \nmid 26921090$. The survivor totals are deduplicated within their respective campaigns. The second statement combines the second campaign with Corollary 3.3. □

| campaign | pairs $(p_1, p_2)$ | window survivors | Fermat survivors | solutions | wall time |
|---|---|---|---|---|---|
| all $n \le 10^{30}$ | 230,612,582,193 | 1,904 | 1 | 0 | 87 min |
| all $n$, $p_1 \le 500{,}000$ | 86,517,004,140 | 1,185 | 1 | 0 | 146 min |

TABLE 2. The four-prime enumeration (Theorem 6.1).

## 7. No Williams number below $10^{24}$

**Theorem 7.1.** *No Williams number is smaller than* $10^{24}$. *Among the* 308,279,939 *Carmichael numbers below* $10^{24}$, *exactly* 14 *have four prime factors* $p$ *with* $(p+1) \mid (n+1)$ *(seven with* $\omega(n) = 9$, *five with* $\omega(n) = 10$, *two with* $\omega(n) = 11$*), one of them*

$$1550030969055486575800 9 = 37 \cdot 103 \cdot 109 \cdot 127 \cdot 307 \cdot 523 \cdot 613 \cdot 1429 \cdot 2089$$

*(the primes* $37, 109, 1429, 2089$*); none has five or more;* 826 *have exactly three,* 63,231 *exactly two and* 5,807,346 *exactly one. Below* $10^{22}$ *the maximum is three, attained by* 179 *numbers.*

We know of no larger published bound that applies without restriction on the number of prime factors: Pinch's $10^{16}$ [20] is restricted to an odd number of prime factors; McIntosh and Dipra [18] give $10^{24}$ for four prime factors (indeed for Carmichael numbers with four prime factors whose largest prime $s$ satisfies $(s+1) \mid (N+1)$), note that Feitsma's list of base-2 pseudoprimes below $2^{64}$ contains no Williams number, and report a Pinch-style search for $4 \le d \le 13$ prime factors "for various values of $X$ between $10^{20}$ and $10^{35}$" without stating the bound reached for each $d$.

**Input data.** Carmichael numbers have been tabulated completely to $10^{21}$ by Pinch [20, 21], to $10^{22}$ by Goutier [12] and to $10^{24}$ by Shallue and Webster [23, 24], whose list is distributed by Stamm-Wilbrandt as a PARI/GP binary vector [25]. A Williams number is in particular Carmichael, so it suffices to test every tabulated Carmichael number for the Lucas–Carmichael property. This is a factor-and-test pass over an external complete tabulation, not an independent tabulation of the Carmichael numbers; the tabulation's completeness is Shallue and Webster's result, which we check for consistency as described below.

**Per-entry verification.** Each $n$ was factored and verified by us rather than read from the tabulators' flags: Miller–Rabin square-root-of-unity splitting (for a Carmichael number, bases that are strong liars yield no split, so bases are tried until a witness produces a nontrivial square root of 1; general factorisation is the fallback), then the factorisation was checked to be squarefree with product $n$ and to satisfy Korselt's criterion, and the number of primes $p$ with $(p+1) \mid (n+1)$ was recorded. Processing is line-based so that no entry can be skipped at a buffer boundary.

**Integrity checks.** The counts below $10^{22}$, $10^{23}$ and $10^{24}$ (49,679,870; 123,381,982; 308,279,939) equal the published ones [24]; the list is strictly increasing and begins $561, 1105, 1729, 2465, 2821$; our per-$\omega$ tallies agree with independently published counts, 1,494,738 three-prime Carmichael numbers below $10^{24}$ (OEIS A132195, [8]) and 6,529,921 eleven-prime ones (OEIS A299711); below $10^{22}$ all eleven per-$\omega$ counts agree with Goutier's tally, and the 179 three-match numbers below $10^{22}$ agree with OEIS A329948 (Suteu; b-file by Eldar). The converse direction was also checked: none of the 129,125 Lucas–Carmichael numbers below $10^{15}$ tabulated by Suteu (OEIS A006972) is Carmichael. Table 3 gives the distribution below $10^{24}$. The scan was run twice: the first run recorded the per-$\omega$ counts, the three-match count and one number attaining the maximum; the second (checkpointed) run recorded the full histogram of matching primes by $\omega(n)$ and is certified by a script that asserts a contiguous set of chunks, the total 308,279,939, histogram sum equal to the total and zero verification failures; the two runs agree on every shared statistic.

| $\omega(n)$ | count | $\omega(n)$ | count | $\omega(n)$ | count |
|---|---|---|---|---|---|
| 3 | 1,494,738 | 7 | 62,338,997 | 11 | 6,529,921 |
| 4 | 548,265 | 8 | 99,469,866 | 12 | 547,590 |
| 5 | 3,429,390 | 9 | 81,041,495 | 13 | 15,328 |
| 6 | 19,665,194 | 10 | 33,199,094 | 14 | 61 |

TABLE 3. Carmichael numbers below $10^{24}$ by number of prime factors, as recounted by our scan (total 308,279,939).

**Corollary 7.2.** (a) *A counterexample to Agrawal's conjecture of the Lenstra–Pomerance type (a Williams number all of whose primes are* $\equiv 3 \pmod{80}$*) exceeds* $10^{24}$.

(b) *Any counterexample to the Baillie–PSW test produced by Pomerance's construction* [22] *exceeds* $10^{24}$*, because that construction produces a Williams number.*

Statement (b) adds little to the known absence of Baillie–PSW counterexamples below $2^{64} \approx 1.8 \times 10^{19}$ [11], but it bounds a construction rather than a tested range.

*Remark* 7.3 (Howe's order-2 Carmichael numbers). By Howe's criterion [14], a squarefree $n$ is a Carmichael number of order 2 iff for each $p \mid n$, $n \equiv 1$ or $n \equiv p \pmod{p^2-1}$. For a Williams number the first option is impossible ($n \equiv p \pmod{L_p}$ and $n \equiv 1 \pmod{p^2-1}$ would give $L_p \mid p-1$), so a Williams number is of order 2 iff $n \equiv p \pmod{p^2-1}$ for all $p \mid n$, i.e. iff $(n-p)/L_p$ is even for every $p \mid n$. Tabulations of order-2 Carmichael numbers therefore bound one 2-adic type of Williams number; we have not pursued this.

## 8. Discussion

A heuristic for four prime factors (speculative). Condition on Theorem 3.1(a). For $p \equiv 1 \pmod{12}$ the condition $n/p \equiv 1 \pmod{L_p}$ is automatic modulo 12; treat the remaining condition as an event of probability $24/(p^2-1) \approx 24/p^2$, independently for the four primes (the four events are in fact arithmetically dependent, and no error term is claimed). Summing $\prod_i 24/p_i^2$ over quadruples of primes $\equiv 1 \pmod{12}$ with $p_2^2 \le 24p_1^3+1$ and $p_2 < q < r < 2p_1p_2$ gives about $6.3 \times 10^{-7}$ (script `agrawal/k4_heuristic.py`), of which about 96% comes from $p_1 = 13$; the contribution of $p_1 > X$ decays like $X^{-4}$ up to logarithms. On this model four-prime Williams numbers are not expected, and Theorem 6.1 is consistent with the prediction.

Open problems.

(1) Is $k = 4$ impossible? By Theorem 3.1, fixing $p_1$ leaves a finite search over $(p_2, q, r)$; the integer $\delta = des - 8p_1^3 = 8p_1^3(1/\Pi - 1)$ satisfies $\delta \geq 1$ [18, proof of Thm. 2] and ties the sizes of $p_2, q, r$ to $p_1$ but does not by itself exclude them.
(2) The $k = 5$ case, where both classes modulo 4 are allowed and the analogue of (2) has four factors.
(3) Closing the $m = 4$ polynomial-family stratum with one repeated root (Appendix A).

## 9. Reproducibility

All code is in the repository `https://github.com/razv2025/carmichael-lucas-carmichael`, archived on Zenodo under the concept DOI `https://doi.org/10.5281/zenodo.23886954` (which resolves to the latest archived version; this paper corresponds to tag `v1.1`); the certification records state the SHA-256 of the enumerator source and binary they certify. Inventory of the computations and their verification status: (i) the two four-prime campaigns were run twice, certified as described in §5, and the certifications were repeated by an external reviewer from the tagged source; (ii) the $10^{24}$ scan was run twice (the second run with the full match histogram and a certification record), with the integrity checks of §7 against externally published counts, and its input is identified by SHA-256; (iii) the relaxed-problem validation to $10^{12}$ and the self-tests are archived and re-runnable (`make test`, under a minute); (iv) the polynomial-family searches of Appendix A are archived with their command lines. The four-prime enumeration (`clc/k4/`) consists of the C program, its brute-force reference for the relaxed problem, the Python re-verifier, the per-block logs and survivor dumps of both campaigns, the certification records and `make test`; the $10^{24}$ scan (`clc/`) consists of the download–decode–verify–scan pipeline, the completeness report and the result files. The code was developed with AI assistance; see the declaration below.

## Appendix A. Polynomial families

Carmichael numbers are constructed in two ways: by polynomial families in the manner of Chernick [7], $n(k) = (6k+1)(12k+1)(18k+1)$, where the divisibilities are polynomial identities; and by subset products in the manner of Erdős and Alford–Granville–Pomerance [10, 3, 15]. This appendix shows that the first construction cannot be made two-sided in its simplest forms. *The theorems concern families whose divisibilities hold identically in the parameter $k$; they say nothing about families that are Williams numbers only for $k$ in some residue classes, nor about sporadic values.* A measurement of the second construction is reported, with its limitations, in the exploratory note [26]; nothing here depends on it.

Consider $n(k) = \prod_{i=1}^{m}(a_i k + b_i)$ with integers $a_i > 0$, the linear factors pairwise non-proportional, and require $(a_i k + b_i - 1) \mid (n(k) - 1)$ and $(a_i k + b_i + 1) \mid (n(k) + 1)$ in $\mathbb{Q}[k]$. Since a linear polynomial divides $f$ iff $f$ vanishes at its root, this is $n\big((1 - b_i)/a_i\big) = 1$ and $n\big((-1 - b_i)/a_i\big) = -1$. With $D_{ij} = a_i b_j - a_j b_i$ one computes $a_j \frac{1-b_i}{a_i} + b_j = (a_j + D_{ij})/a_i$, so the conditions read, for every $i$,

$$\prod_{j \neq i}(a_j + D_{ij}) = a_i^{m-1} = \prod_{j \neq i}(D_{ij} - a_j). \tag{5}$$

If $g = \gcd(a_1, \ldots, a_m) > 1$, write $a_i = g a_i'$; then $D_{ij} = g D_{ij}'$ and (5) for $(a, b)$ is $g^{m-1}$ times (5) for $(a', b)$, so every family reduces to one with $\gcd(a_i) = 1$ (the same numbers on the sub-progression $k \in g\mathbb{Z}$); the searches below therefore assume $\gcd(a_i) = 1$.

**Theorem A.1** (three factors)**.** *No such family exists for $m = 3$.*

*Proof.* Subtracting the two products in (5) for $i = 1, 2, 3$ gives $a_2D_{13} + a_3D_{12} = 0$, $a_1D_{23} + a_3D_{21} = 0$, $a_1D_{32} + a_2D_{31} = 0$, i.e. $a_2D_{13} = -a_3D_{12}$, $a_1D_{23} = a_3D_{12}$, $a_1D_{23} = -a_2D_{13}$. The identity $a_1D_{23} - a_2D_{13} + a_3D_{12} = 0$ (it is $a \wedge b \wedge a = 0$) becomes $3a_3D_{12} = 0$, so $D_{12} = 0$ and then $D_{13} = D_{23} = 0$: $b$ is proportional to $a$ and the factors are proportional, contrary to hypothesis. □

**Theorem A.2** (uniform offsets)**.** *No such family exists with $b_i = -1$ for all $i$, nor with $b_i = +1$ for all $i$, for any $m \geq 2$.*

*Proof.* If all $b_i = -1$ then $D_{ij} = a_j - a_i$, the second product in (5) is $(-a_i)^{m-1}$, and the first becomes $\prod_{j\neq i}(2a_j - a_i) = a_i^{m-1}$. The $a_i$ are distinct (non-proportionality). Take $i$ with $a_i$ maximal: for $j \neq i$, $0 < a_j < a_i$ gives $|2a_j - a_i| < a_i$, so the product has absolute value $< a_i^{m-1}$. The case $b_i = +1$ is the mirror image with $\prod_{j\neq i}(a_i - 2a_j)$. □

**Theorem A.3** (generic position)**.** *Put $s_i = (b_i - 1)/a_i$ and $t_i = (b_i + 1)/a_i$. If the $s_i$ are pairwise distinct and the $t_i$ are pairwise distinct, no such family exists, for any $m \geq 2$.*

*Proof.* Let $G(\tau) = n(-\tau)$, of degree $m$ with leading coefficient $c = (-1)^m \prod a_i$. The hypotheses say $G(s_i) = 1$ and $G(t_i) = -1$. With $m$ distinct roots each, $G - 1 = c\prod(\tau - s_i)$ and $G + 1 = c\prod(\tau - t_i)$. Subtracting, $2 = c\big[\prod(\tau - t_i) - \prod(\tau - s_i)\big]$ identically in $\tau$. The coefficient of $\tau^{m-1}$ on the right is $-c\sum(t_i - s_i) = -c\sum 2/a_i \neq 0$, while on the left it is 0. □

**Corollary A.4.** *Every such family has two indices $i \neq j$ with $p_j - 1 = \frac{a_j}{a_i}(p_i - 1)$ or $p_j + 1 = \frac{a_j}{a_i}(p_i + 1)$ as polynomial identities.*

The strata left open by Theorem A.3 were searched exhaustively using the fact that each factor in (5) divides $a_i^{m-1}$: no family exists with $m = 4$ and all $a_i \leq 500$, or with $m = 5$ and $a_i \leq 150$; Theorem A.2 was also confirmed by search for $m = 5, 7, 9, 11$ up to $a_i \leq 20000, 5000, 3000, 1500$. The outputs of these runs, with their command lines, are archived in `agrawal/results/`. Theorem A.1 is the parametric shadow of Proposition 2.3.

## Declaration of generative AI and AI-assisted technologies

Two AI systems were used, under the author's direction. Claude (Anthropic) drafted and checked the proofs, wrote the code (the enumerator, the scanners, the certification and test scripts), drafted this text, performed the literature search, and ran adversarial checks of the proofs and of the numerical claims. OpenAI Codex, operated separately from the authoring system, produced three rounds of review comments on successive drafts, identified the prior Williams-number literature, and repeated both production certifications from a clean tagged checkout; its findings led to the corrections recorded in the repository. The author reconstructed and checked every proof and every claim, reviewed the code, and accepts full responsibility for the content. No independent human review of this manuscript had been completed at the time of writing; AI checks are not a substitute for it.

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